

20260415EB_MN (RNA)

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Status: In progress

Entry Created On: 12 May 2026 21:54:49 UTC

Entry Last Modified: 22 May 2026 18:56:56 UTC

Export Generated On: 01 Sep 2026 18:33:45 UTC

2026年4月15日 星期三

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 1 of 6-wells with 1 well of 6-well cell-repellent plates (2 wells for one strain)
keep some iPSCs

2026年4月17日 星期五

T2 -> T3

2026年4月19日 星期日

T3 -> T4 (~14 mL for over-weekend)

2026年4月22日 星期三

T4 -> T5

2026年4月24日 星期五

T5 -> T6

2026年4月27日 星期一

T6 -> T6 (~14mL for over-weekend)

2026年4月29日 星期三

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info			^
	A	B	
1	cell	passage	
2	iPSC1000		
3	iPSC1066		
4	iPSC1108		

Table1			^
	A	B	
1	per well	volume (mL)	
2	trypsin	1	
3	HI-FBS/CTWM	2	

Table3										/
	A	B	C	D	E	F	G	H	I	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	4.26	99	7	29.82	A	96	60	36	507	
3					B	24	250	16	939	
4					C,D	6	1972	12	the rest	

Well1					^
	1	2	3	4	
A	1000 (1.25M/well)				
B	1066 (680K/well)				
C	1108 (1.025M/well)				

- 2026年5月1日 星期五
- T7 -> T8
- 2026年5月3日 星期日
- 1/2 T8 -> 1/2 T8
- 2026年5月5日 星期二
- 1/2 T8 -> 1/2 T8
- 2026年5月7日 星期四
- 1/2 T8 -> 1/2 T8
- 2026年5月9日 星期六
- 1/2 T8 -> 1/2 T8
- 2026年5月11日 星期一
- 1/2 T8 -> 1/2 T8
- 2026年5月13日 星期三

Treat MMS for 1h and collect RNA

MMS might be hydrolyzed at this point since I prepared the media around noon.

Well2				
	1	2	3	4
A	0mM MMS	1.5mM MMS	0mM MMS	1.5mM MMS
B	0mM MMS	1.5mM MMS	0mM MMS	1.5mM MMS
C	0mM MMS	1.5mM MMS		

Treatment time		
	A	B
1		12 well plate for 20260415MN
2	treat	21:45-50
3	collect	22:45-50

1. Add 500uL Trizol per well
2. Combine Column 1 and 3 together for iPSC1000 and iPSC1066